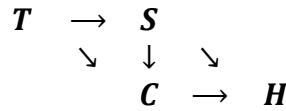


1. Confounding and DAGs

a. DAG



b. S is confounder.

- It is not mediator because it does not lie on the causal pathway from C to H .
- It is common cause of both C and H .

c. There are two backdoor paths:

$$\text{Path 1: } C \leftarrow S \rightarrow H, \quad \text{Path 2: } C \leftarrow T \rightarrow S \rightarrow H$$

To get unbiased estimate, all backdoor paths must be blocked.

- Controlling for T alone is not enough: It blocks Path 2 but leaves the confounding path $C \leftarrow S \rightarrow H$ open.
- Controlling for S alone is sufficient: It blocks Path 1 and it also blocks Path 2.

Therefore, the minimal controlling set is $\{S\}$.

2. Counterfactual outcome framework

- The fundamental problem of causal inference is that, for any individual unit, we can observe only one of their counterfactual outcomes. For example, we may see how many times user 1 clicks with new button $Y_i(1)$, or how many times they click with the old button $Y_i(0)$, but we can never observe both for the same user at the same time.
- Individual-level treatment effect

i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$	A	Y
1	12	8	4	1	12
2	7	5	2	1	7
3	10	11	-1	1	10
4	14	9	5	1	14
5	8	6	2	1	8
6	9	4	5	0	4
7	11	7	4	0	7
8	5	3	2	0	3
9	13	10	3	0	10
10	6	6	0	0	6

c. Average Treatment Effect (ATE)

$$E[Y(1) - Y(0)] = \frac{4 + 2 - 1 + 5 + 2 + 5 + 4 + 2 + 3}{10} = \frac{26}{10} = 2.6$$

d. Observed difference in means

$$E(Y|A = 1) - E(Y|A = 0) = \frac{(12 + 7 + 10 + 14 + 8) - (4 + 7 + 3 + 10 + 6)}{5} = \frac{21}{5} = 4.2$$

is much larger than ATE. This discrepancy is due to **selection bias**, since the assignment of users to treatment and control groups was not random. $E[Y(a)]$ cannot be identified as $E(Y|A = a)$.